

CS3081 · INTELLIGENT AGENTS UNDER ADVERSARIAL CONDITIONS

Intelligent Multi-Agent Rainwater Management for Urban Flood Mitigation

Heavy rainfall in cities like Jeddah causes rapid street flooding and disrupted transport — while rainwater that could be stored is lost to uncontrolled runoff.

We propose a multi-agent system that monitors rainfall, coordinates collection and storage, and reroutes water in real time as conditions change.

Amira Atiah (S24109222) | Anosha Kanwal (S24109479) | Ward Shakir (S24109277)

RESEARCH QUESTION

Can agents reason correctly when the environment is incomplete and changing?

Our agents will face genuine adversarial conditions during a rainfall event, not just clean simulated data:

Incomplete knowledge — blocked routes

Noisy observations — unreliable sensors

Multi-agent conflict — competing storage demand

ARCHITECTURE

Five specialized agents, working together



Rainfall Monitoring



Water Collection



Route / Drainage



Storage Management



Emergency Response

Agents communicate and negotiate — e.g. if one reservoir nears capacity, the system redirects incoming water to another location.

METHODS

Classical AI techniques, not a CV/ML pipeline

A* Search

Route planning around blocked or high-risk roads

Knowledge Representation

Environmental rules and constraints

Probabilistic Reasoning

Handling uncertain rainfall & sensor data

Planning

Managing storage allocation over time

EXAMPLE SCENARIO

How the agents adapt mid-event

- 1 Monitoring agent detects rising rainfall
- 2 Areas begin accumulating water
- 3 Collection agent identifies available points
- 4 Route agent finds paths, avoiding blocked roads
- 5 Storage agent checks reservoir capacity
- 6 A sensor fails or a route closes unexpectedly
- 7 Agents replan and redirect water accordingly

EVALUATION

What we'll measure

Water collected vs. lost

Runoff reduction effectiveness

Flooded area extent

Urban disruption avoided

Storage utilization

Efficiency of allocation

Adaptation success rate

Response to unexpected events

TEAM & PLAN

Team of 3 — ready to build

We'll implement the simulation incrementally: environment design first, then agent behaviors, then the adversarial scenarios (noisy sensors, blocked routes, storage conflict), followed by evaluation and the report.

Goal: a working, visual simulation that clearly demonstrates multi-agent coordination under uncertainty — directly answering the course's core question.